

VARIABLE

Act as containers that store data or information

Example

```
Int a = 20;
```

Int – data type

A – variable name

20 – value

20 store inside the variable name

Types

1. Local
2. Instant
3. Static

Type Casting

Converting one data types into another

Example:

```
int a = 10;
```

```
double z = a ( convert integer to decimal).
```

Output:

```
a = 10
```

```
Z = 10.0
```

Usage

Perform operation between two different data types and convert smaller to larger data types

Types

- Widening casting → Automatic
- Narrowing Casting → Manual

Widening casting

Other names implicit and upcasting

Its work automatically (no syntax needed)

Example

```
Int x =100;
```

```
Double y=x;
```

Narrowing Casting

Other names explicit and down casting

Convert large to small

Example

```
double z = 100;
```

```
int x = (int) z;
```

Operator

An operator is a symbol that performs an operation on variables or values.

Types

1. Arithmetic – add, sub, multiply, divide
2. Assignment- assign the value and check double side as -=, +=, *=, /=
3. Relational- Comparison (<, >, ==, !=)
4. Logical- AND OR (&&, ||)
5. Unary - single operator (a++ memory save/assigned value print, ++a changed value)
6. Bitwise operator
7. Ternary operator – (syntax- variable = (condition) ? valueIfTrue : valueIfFalse;)

User Input

To take user input, need to use scanner class

Example program:

```
import java.util.Scanner; // 1. Import the Scanner class
```

```
public class UserInputExample {
```

```
    public static void main(String[] args) {
```

```
        // 2. Create a Scanner object to read keyboard input
```

```
        Scanner scanner = new Scanner(System.in);
```

```
        // 3. Prompt the user and read a String (Text)
```

```
        System.out.print("Enter your name: ");
```

```
        String name = scanner.nextLine();
```

```

// 4. Prompt the user and read an Integer (Whole Number)
System.out.print("Enter your age: ");

int age = scanner.nextInt();

// 5. Output the collected data
System.out.println("\nHello " + name + "! You are " + age + " years old.");

// 6. Close the scanner to free system resources
scanner.close();
}
}

```

Java scanner input

Method	Description
<code>nextBoolean()</code>	Reads a <code>boolean</code> value from the user
<code>nextByte()</code>	Reads a <code>byte</code> value from the user
<code>nextDouble()</code>	Reads a <code>double</code> value from the user
<code>nextFloat()</code>	Reads a <code>float</code> value from the user
<code>nextInt()</code>	Reads an <code>int</code> value from the user
<code>nextLine()</code>	Reads a <code>String</code> value from the user
<code>nextLong()</code>	Reads a <code>long</code> value from the user
<code>nextShort()</code>	Reads a <code>short</code> value from the user